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•repulsive force balanced by components of attractive forces between electrons and nuclei.

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3. Without invoking time, what can we find out about changes in molecular structure from simple approximations?

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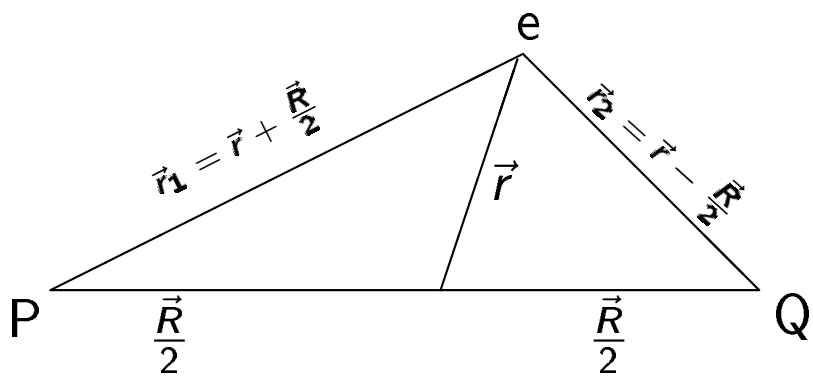
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$$\text{as } R \longrightarrow 0 \quad \mathcal{E} = \frac{e^2}{R} - 2^2 \frac{e^2}{2a_0}$$

wavefunction goes to hydrogenic $Z = 2$ *limit*.

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A good trial function: *LCAO-MO approximation*

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This is called the LCAO-MO approximation (LCAO=Linear combination of atomic orbitals)

consider complete Hamiltonian

$$\hat{H} = \hat{T}_N + \hat{T}_e + V(\vec{r}, \vec{R})$$

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R is now a parameter
and $\mathcal{E}_n(\vec{R})$ *is then a potential energy surface (PES) for nuclear motion*
: also called adiabatic PES

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these terms are small in comparison to 2nd term and can be neglected:
Born-Oppenheimer Approximation

SE, including B.O. approximation:

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Above equation is the eigenvalue problem for the nuclear motion.

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or, dividing both sides by Y,

consider a diatomic molecule again.

Suppose we solve the elctronic problem. In particular, consider ground state energies, $\mathcal{E}_{grd.}(\vec{R}) \equiv \mathcal{E}(\vec{R})$

Turn nuclei loose in this 'potential'.

eigen value problem for nuclei: $\left[-\frac{\hbar^2}{2M}\nabla_R^2 + \mathcal{E}(\vec{R})\right]\phi(\vec{R}) = E\phi(\vec{R})$

$$\text{expand around minimum} : \mathcal{E}(\vec{R}) \approx \mathcal{E}_0 + \frac{1}{2}\left.\frac{\partial^2\mathcal{E}}{\partial R^2}\right|_{R_0}(R-R_0)^2 + ...$$

where $\mathcal{E}_0 = \mathcal{E}(\vec{R})$ *at minimum of internuclear P.E. curve,* $R = R_0$

This is constant in Hamiltonian The second term $= \frac{1}{2}M\omega^2(R-R_0)^2$

$$\phi(\vec{R}) = \frac{1}{R}f(R)Y_j^m(\Omega); \quad \nabla_R^2 = \frac{1}{R}\frac{\partial^2}{\partial R^2}R - \frac{\hat{J}^2}{\hbar^2 R^2}$$

For given j : $\hat{J}^2\phi(\vec{R}) = \hbar^2 j(j+1)\phi(\vec{R})$

$$-\frac{\hbar^2}{2M}\frac{d^2fY}{dR^2} + \left\{\mathcal{E}_0 + \frac{1}{2}M\omega^2(R-R_0)^2 + \frac{\hbar^2 j(j+1)}{2MR^2}\right\}fY = EfY$$

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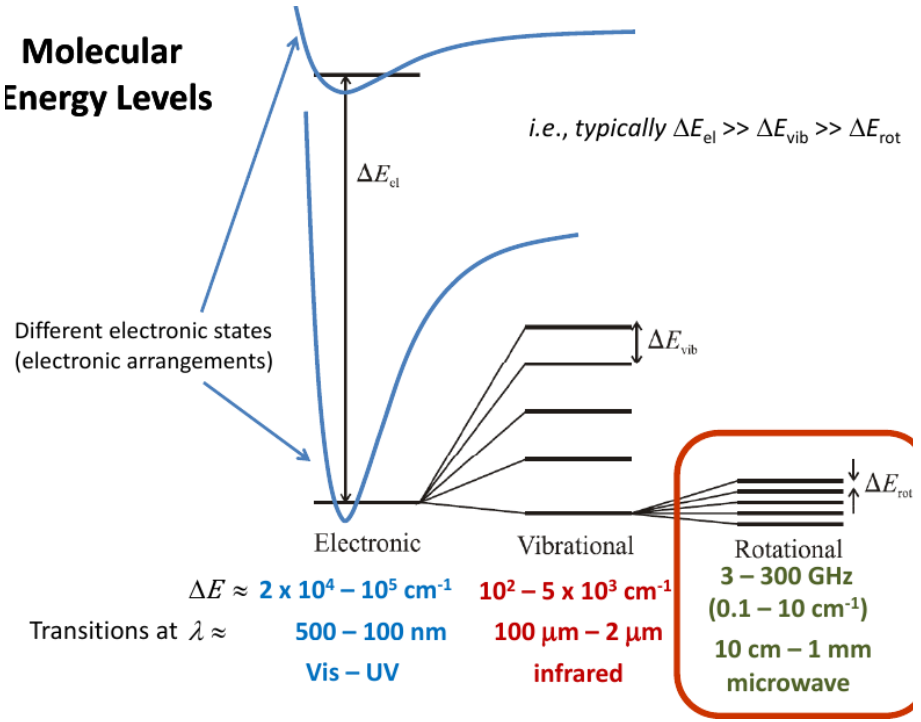
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1st term : electronic, 2nd : vibrational,

and 3rd : rotational part of total energy



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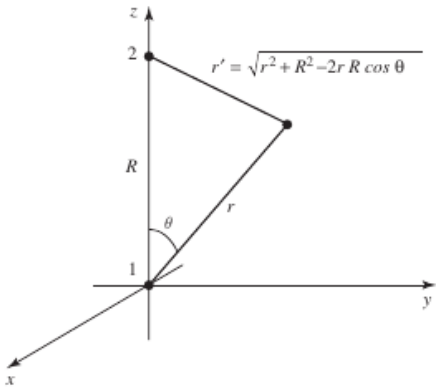
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$$\Longrightarrow \quad |A|^2 = \frac{1}{2(1+I)}$$

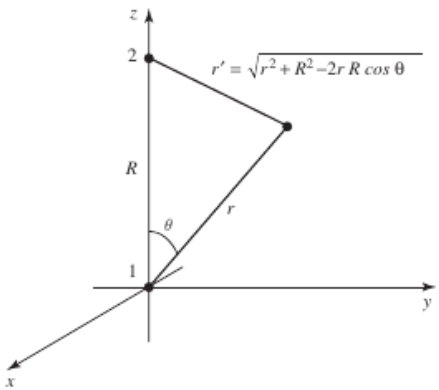
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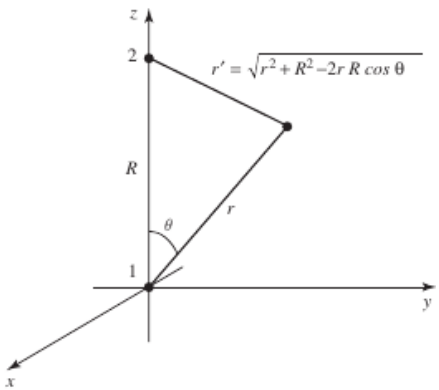


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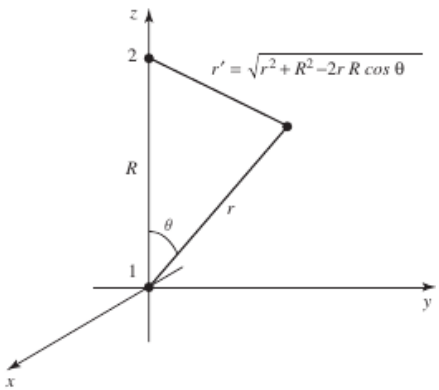


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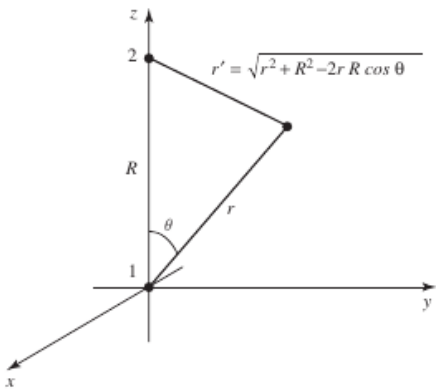
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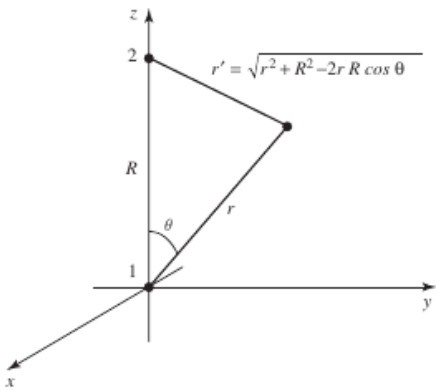
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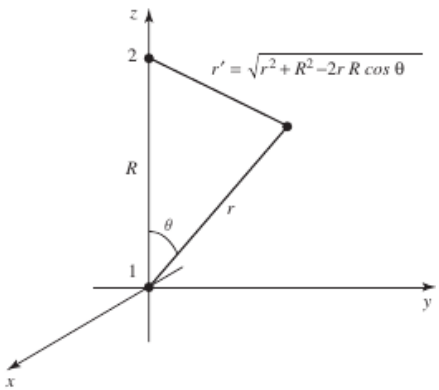
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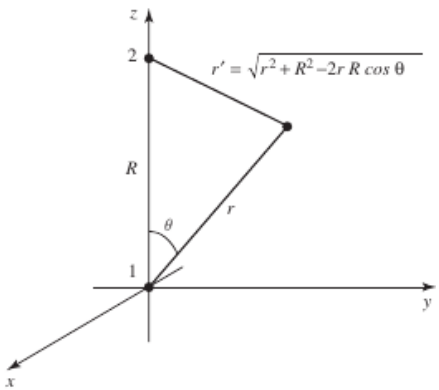
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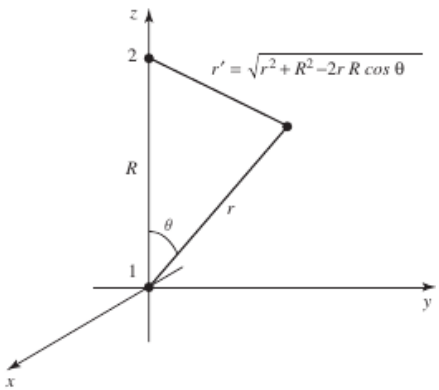
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$$\left(-\frac{\hbar^2}{2m}\nabla^2 - \frac{e^2}{4\pi\epsilon_0}\frac{1}{r}\right)\psi_0(r) = E_1\psi_0(r) \tag{1}$$

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$$\left(-\frac{\hbar^2}{2m}\nabla^2 - \frac{e^2}{4\pi\epsilon_0}\frac{1}{r}\right)\psi_0(r) = E_1\psi_0(r) \tag{1}$$

$E_1 = -13.6$ eV= ground state energy of H atom
and r' in place of r in an integral gives same result

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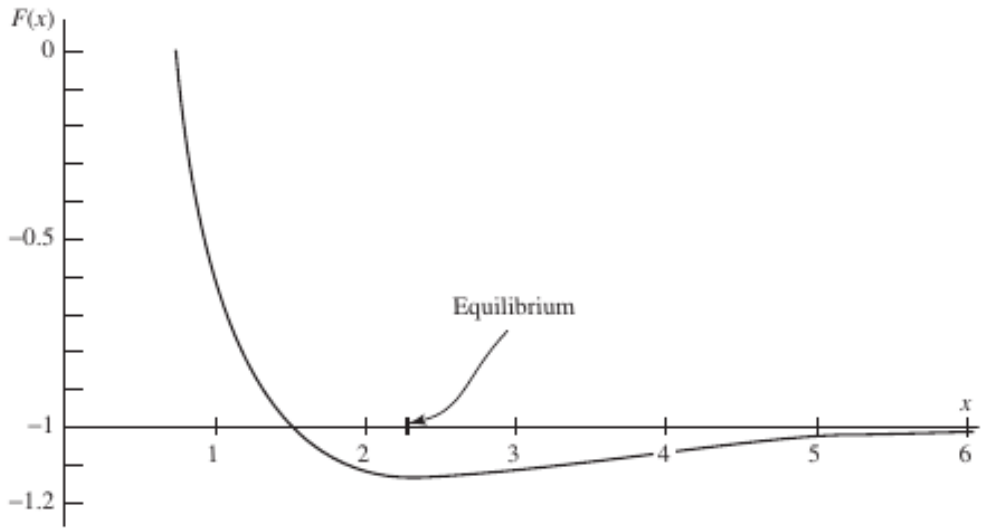
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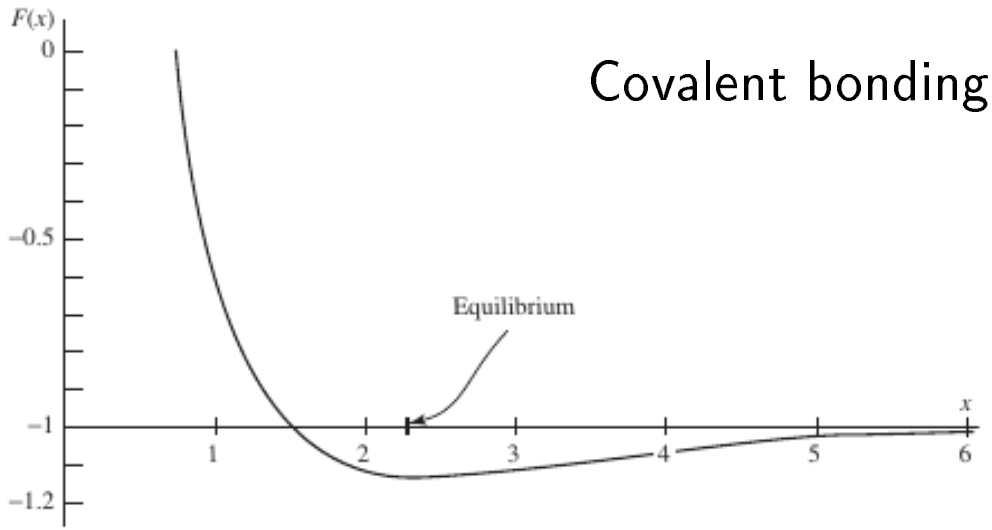
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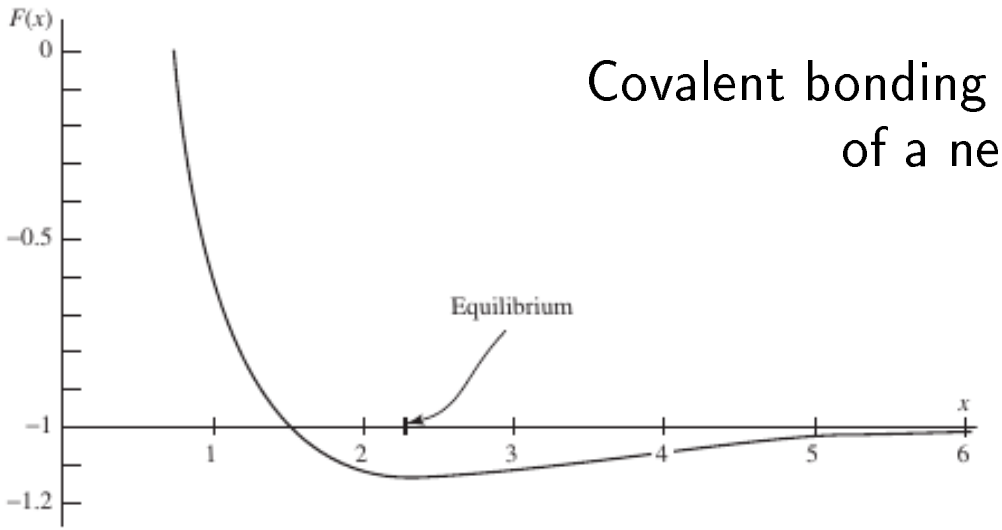
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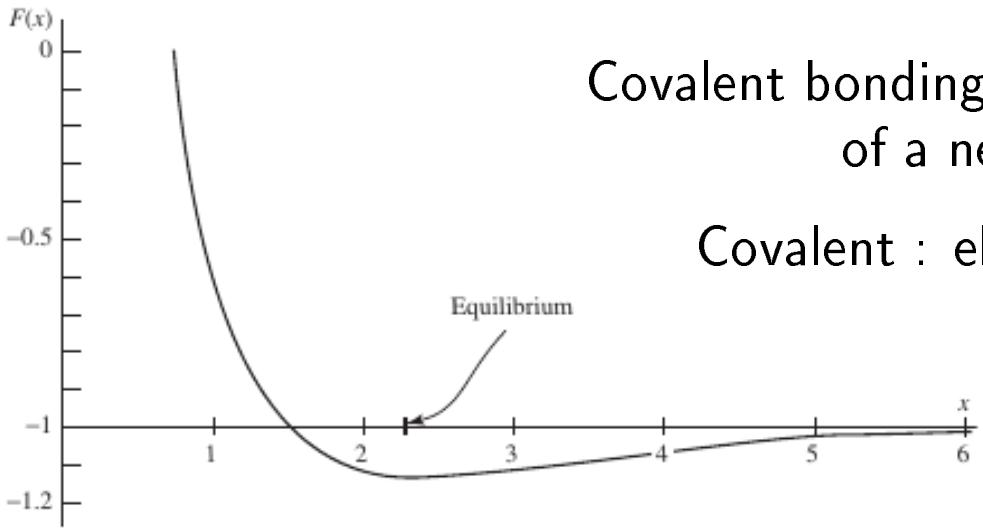
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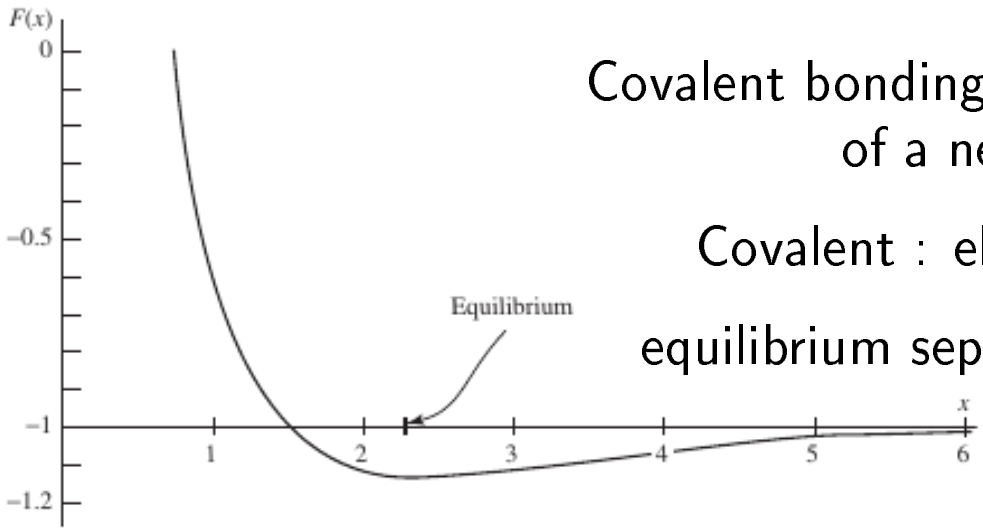
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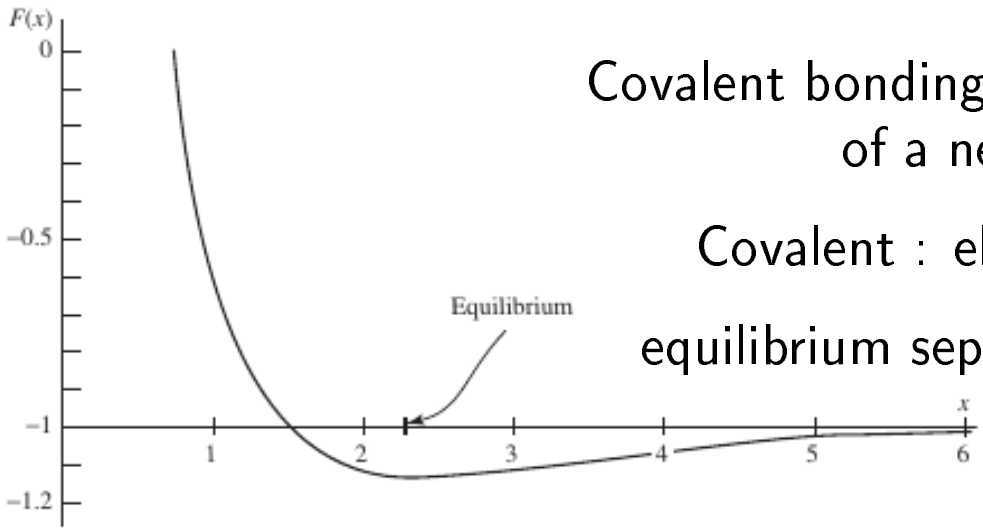
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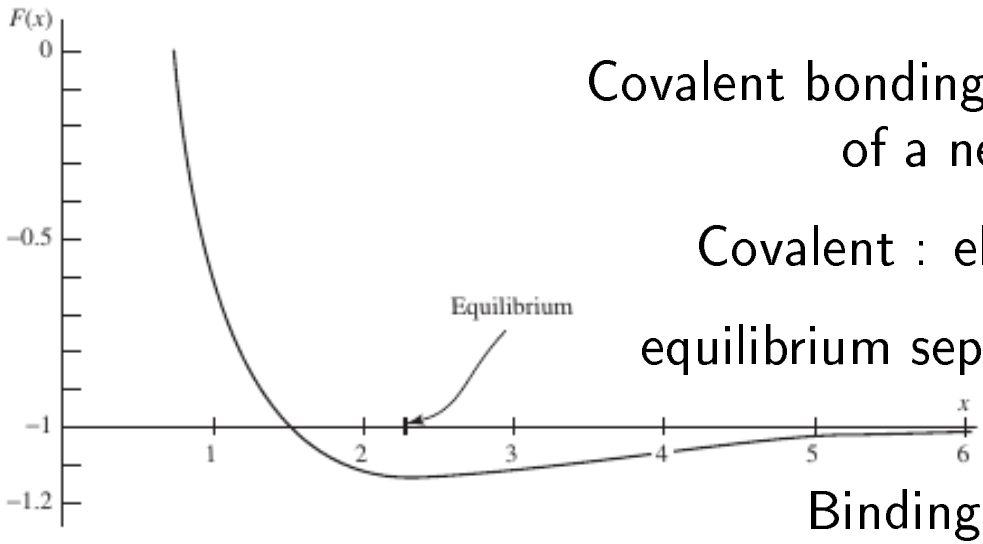
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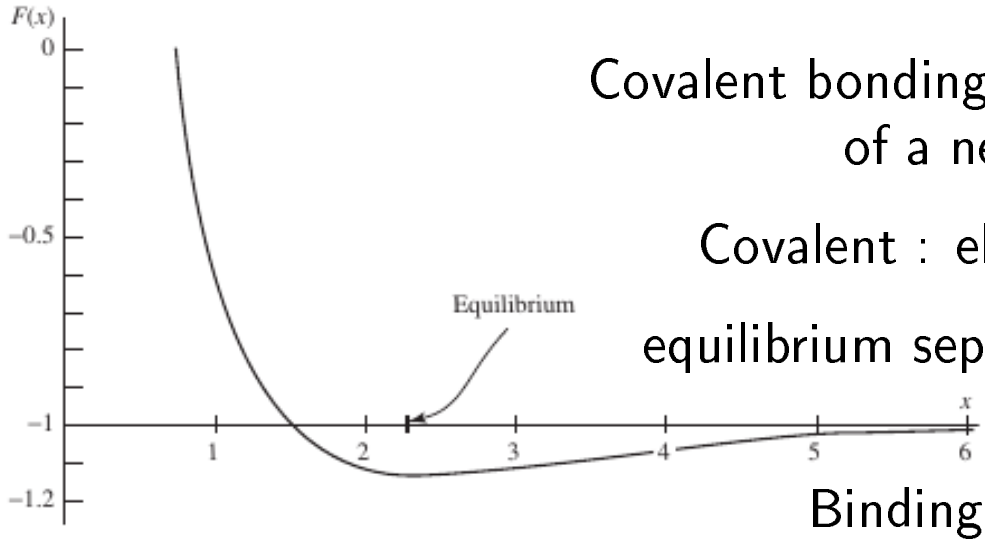
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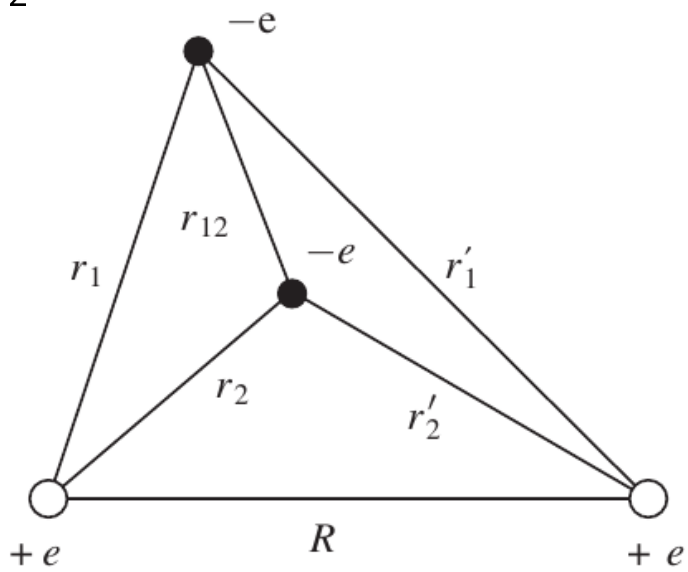
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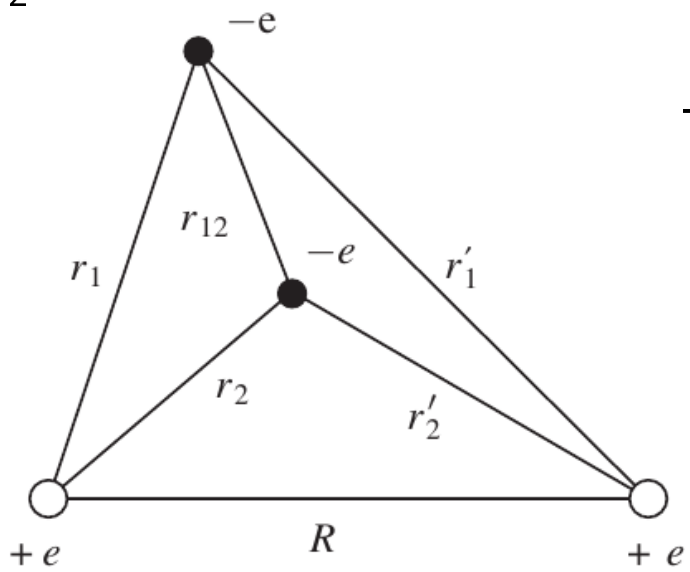
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Binding energy : 1.8 eV, experiment : 2.8 eV

H_2 Molecule

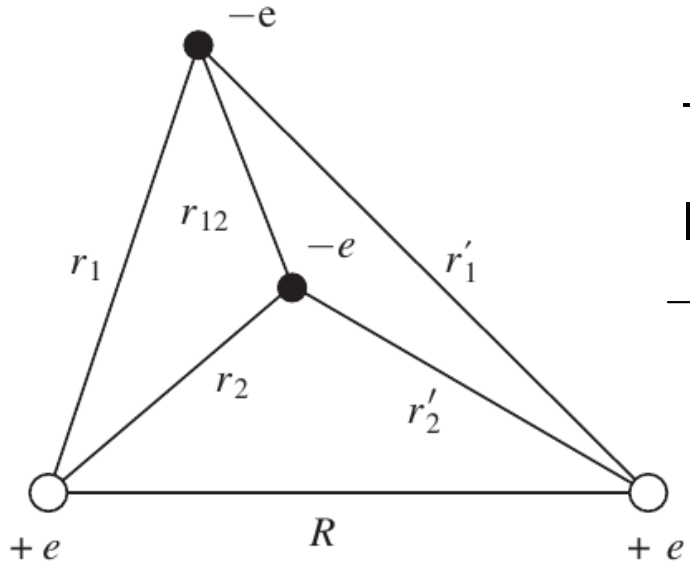


H_2 Molecule



Taking protons to be at rest,

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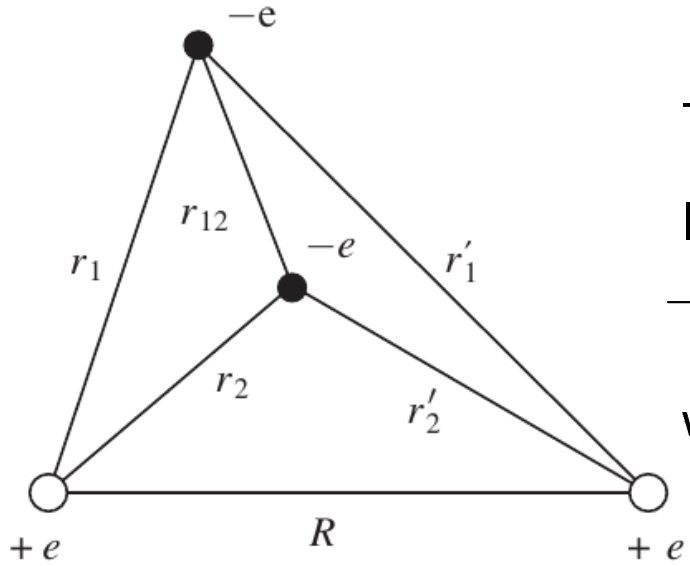


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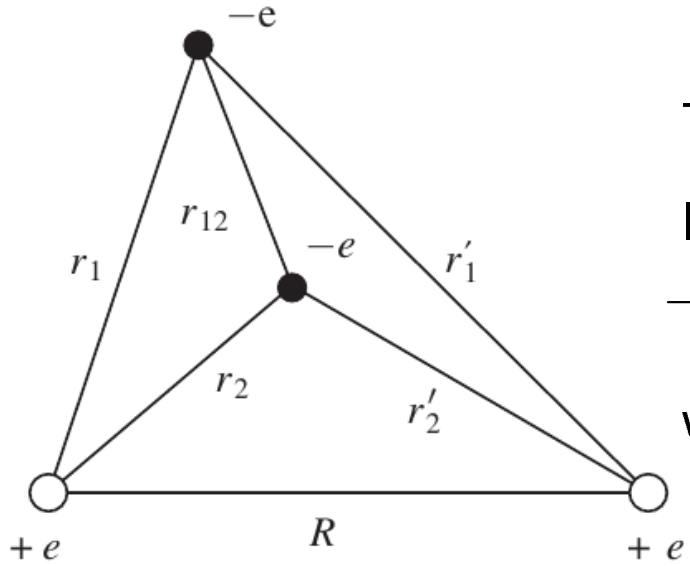
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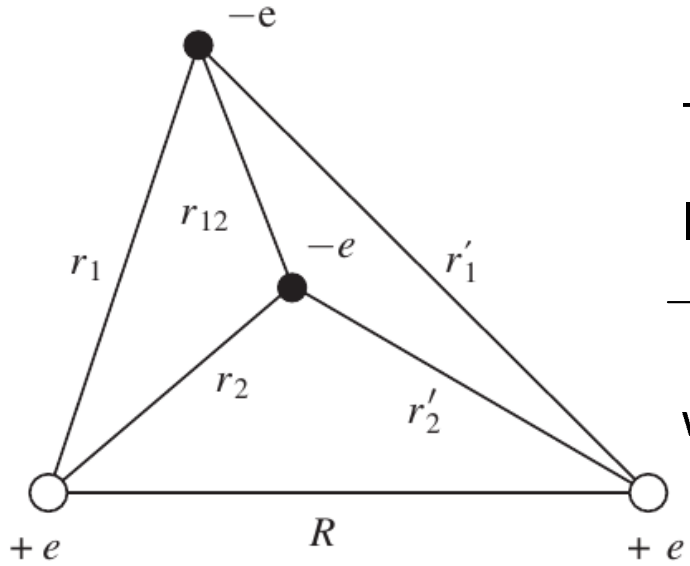
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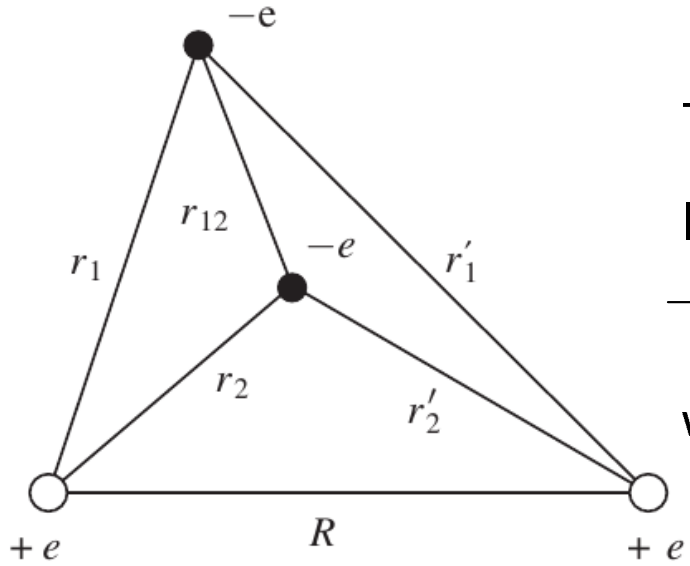
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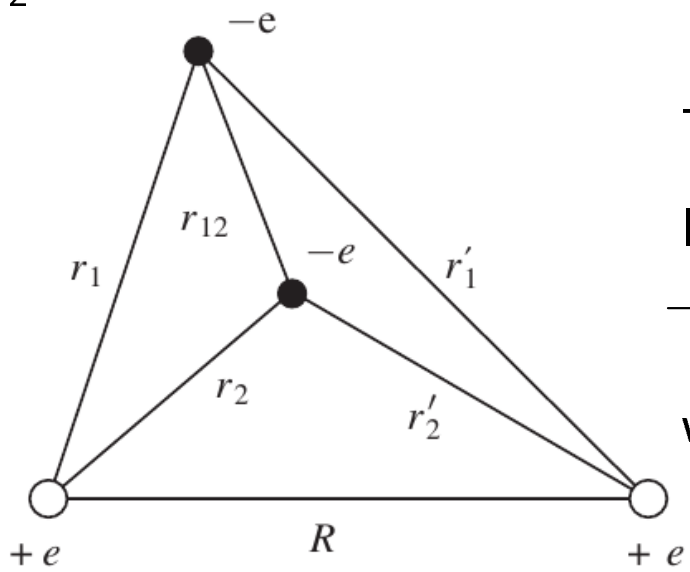
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we could also choose

$\psi_-(r_1, r_2) = A_-[\psi_0(r_1)\psi_0(r_2') - \psi_0(r_1')\psi_0(r_2)] \implies$ spin state symmetric triplet

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For expectation value of the energy, start with kinetic energy of particle 1

Since ψ_0 is the ground state of the hydrogen atom Hamiltonian,

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$= \left(\frac{e^2}{4\pi\epsilon_0 a}\right) A_\pm^2 \left[\int\int \psi_0(r_1)^2 \frac{a}{r_{12}} \psi_0(r_2')^2 d^3r_1 d^3r_2 + \int\int \psi_0(r_1')^2 \frac{a}{r_{12}} \psi_0(r_2)^2 d^3r_1 d^3r_2 \right.$

$\left. \pm 2\int\int \psi_0(r_1)\psi_0(r_1')\frac{a}{r_{12}}\psi_0(r_2)\psi_0(r_2') d^3r_1 d^3r_2\right]$

The first two integrals are equal, as is seen by interchanging the labels 1 and 2

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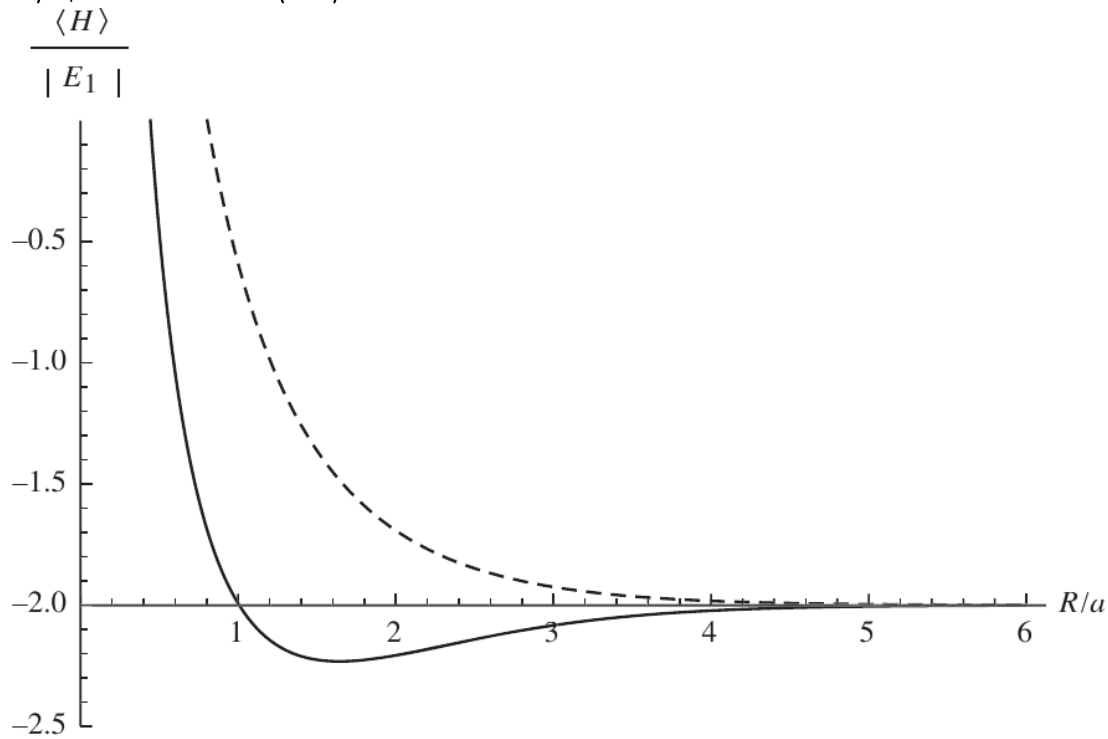
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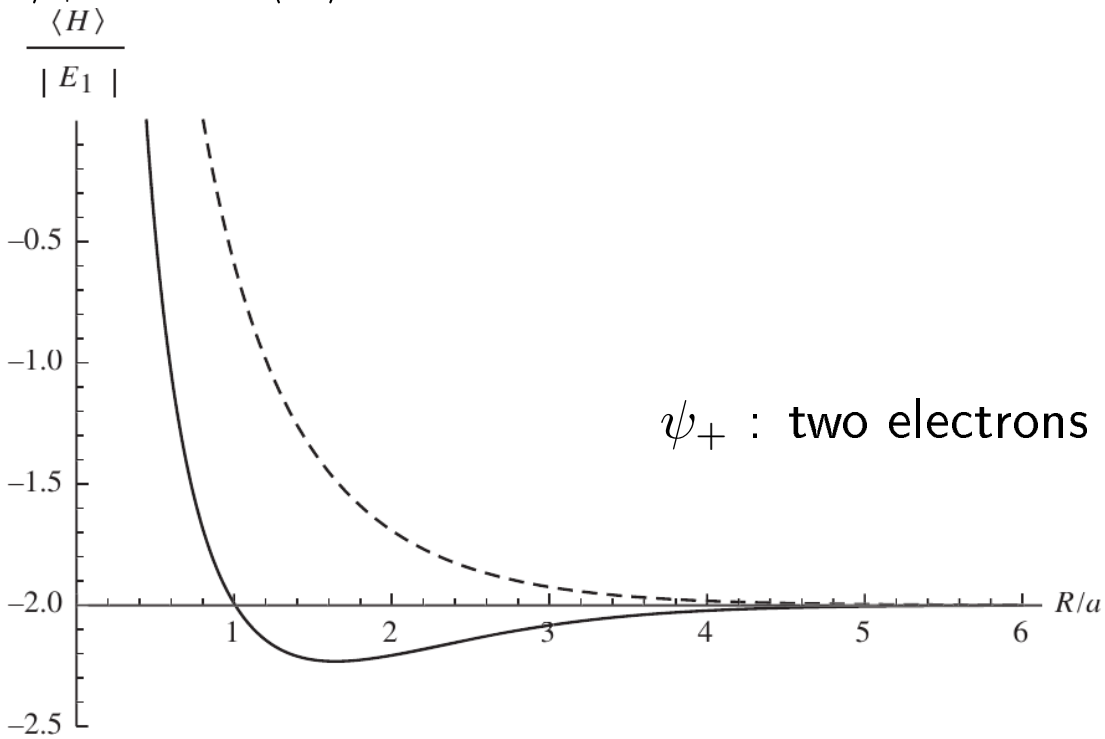
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$\langle H \rangle_\pm = 2E_1 \left[1 - \frac{a}{R} + \frac{2D - D_2 \pm (2IX - X_2)}{1 \pm I^2} \right]$

$\langle H \rangle_+$ versus $\langle H \rangle_-$

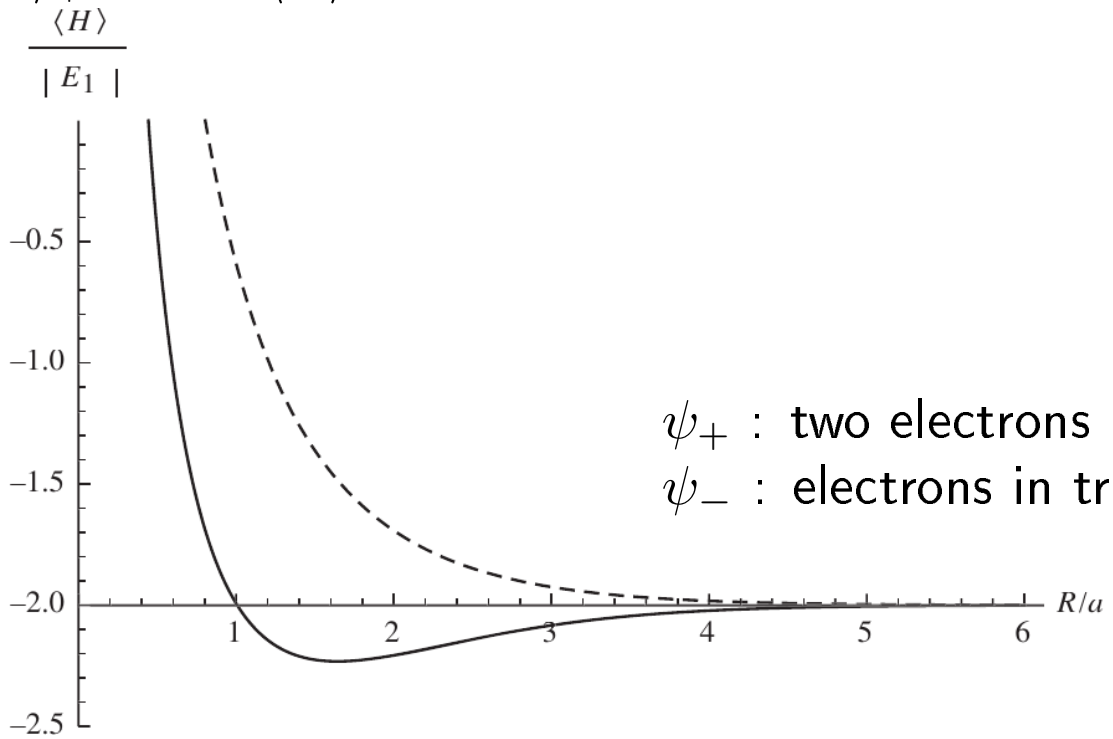


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ψ_+ : two electrons in singlet spin configuration

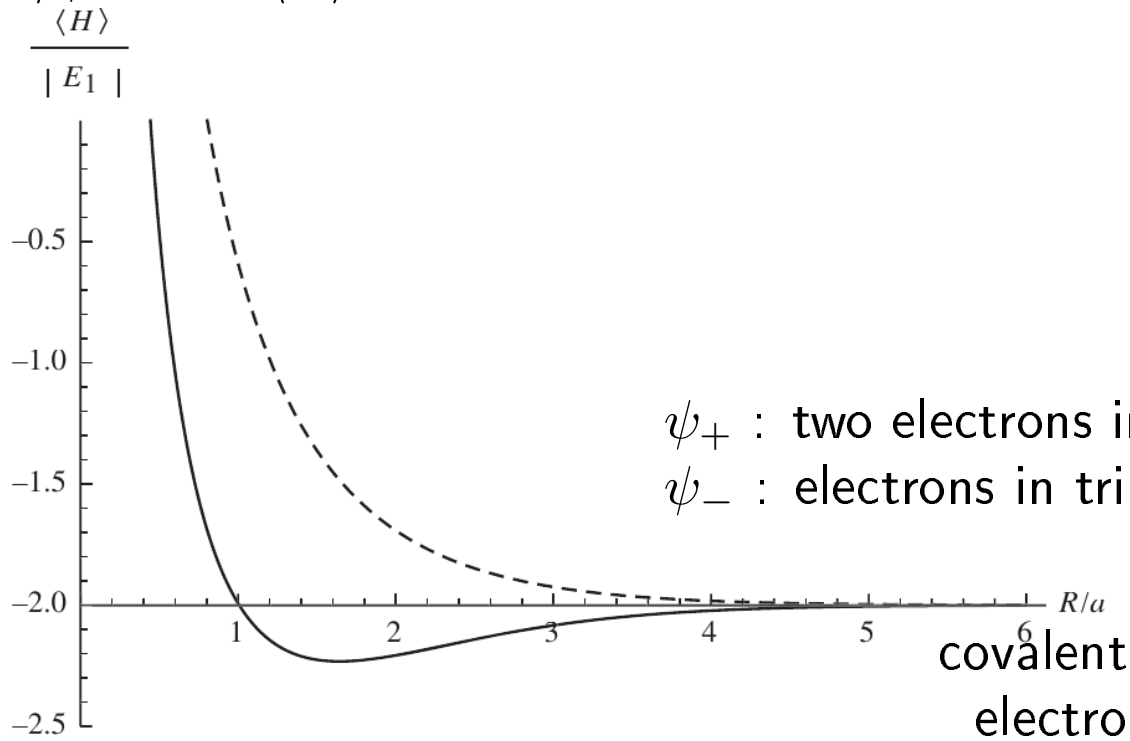
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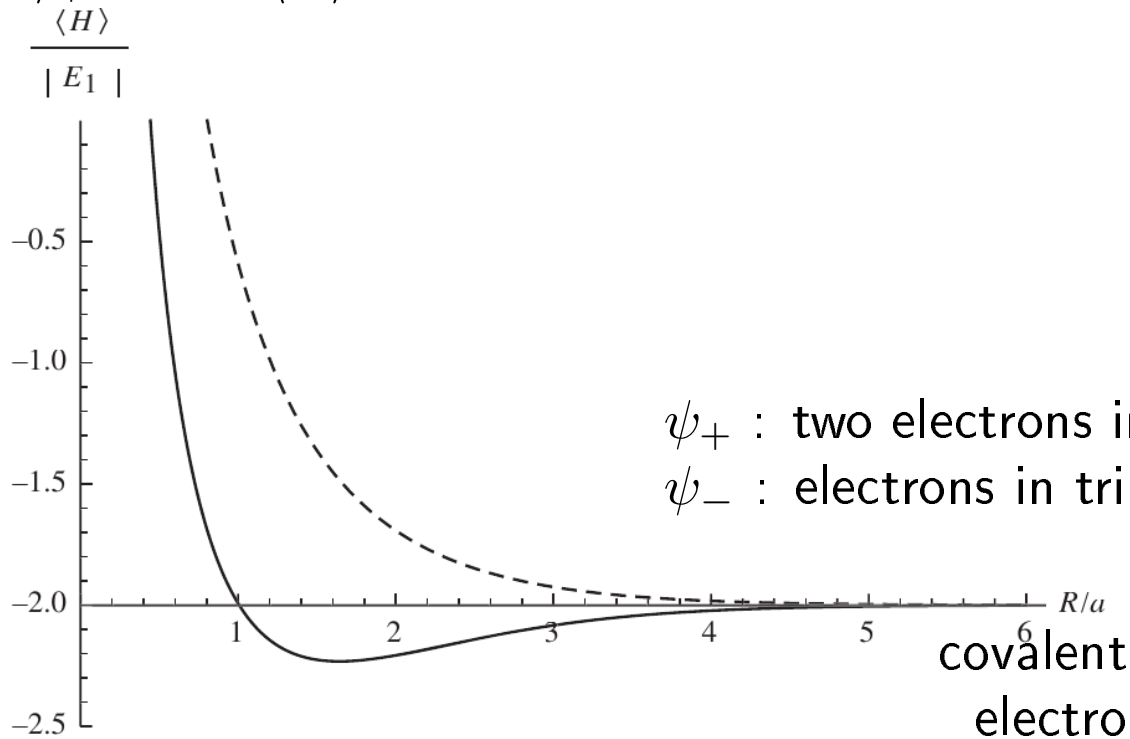


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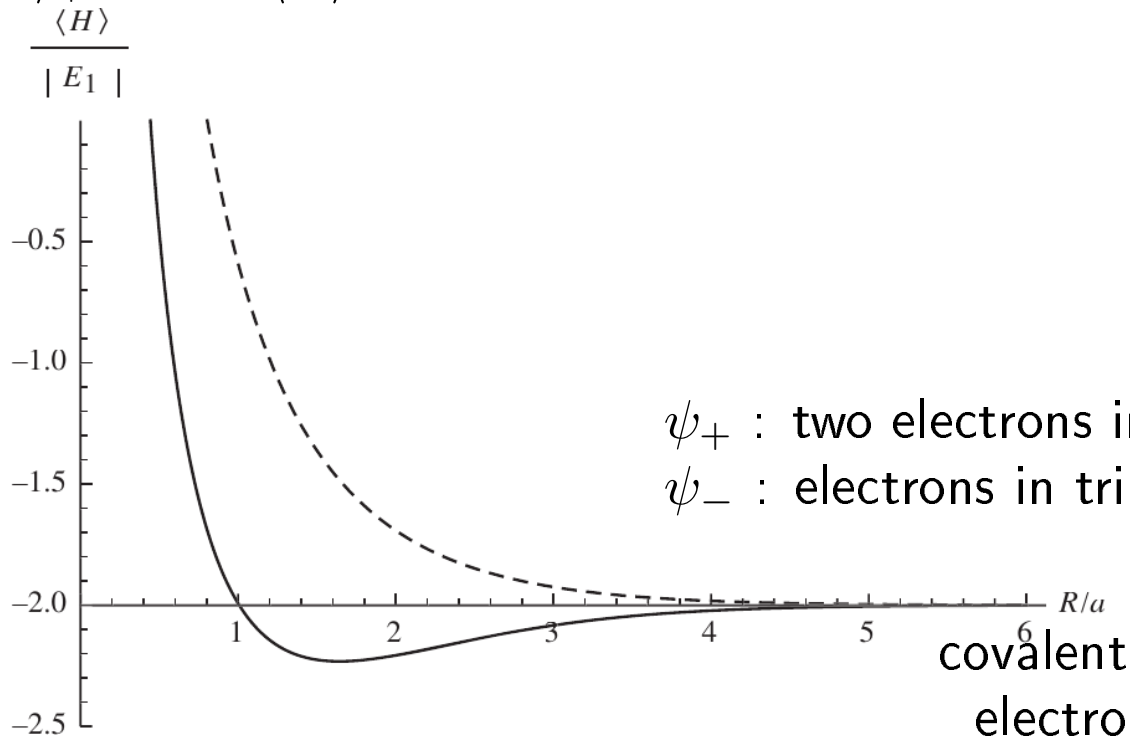


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Locating the minimum on the plot, bond length = 1.64 a.u.;

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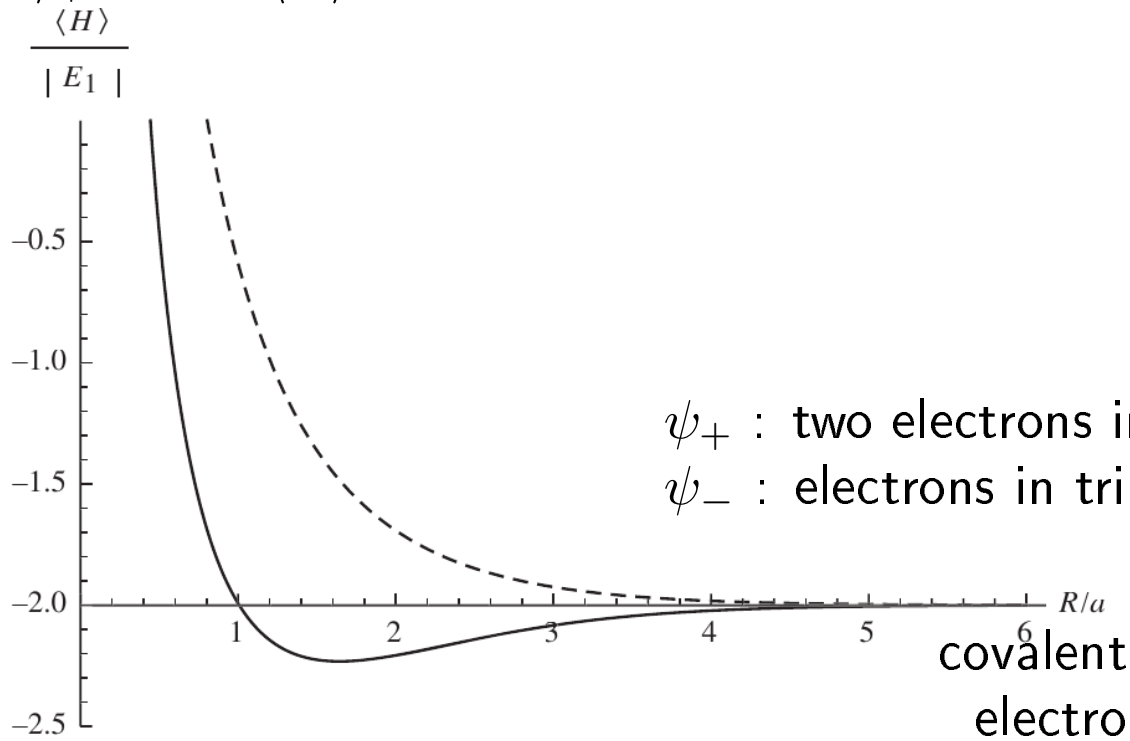


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Locating the minimum on the plot, bond length = 1.64 a.u.; expt. : 1.40 a.u.

$\langle H \rangle_+$ versus $\langle H \rangle_-$



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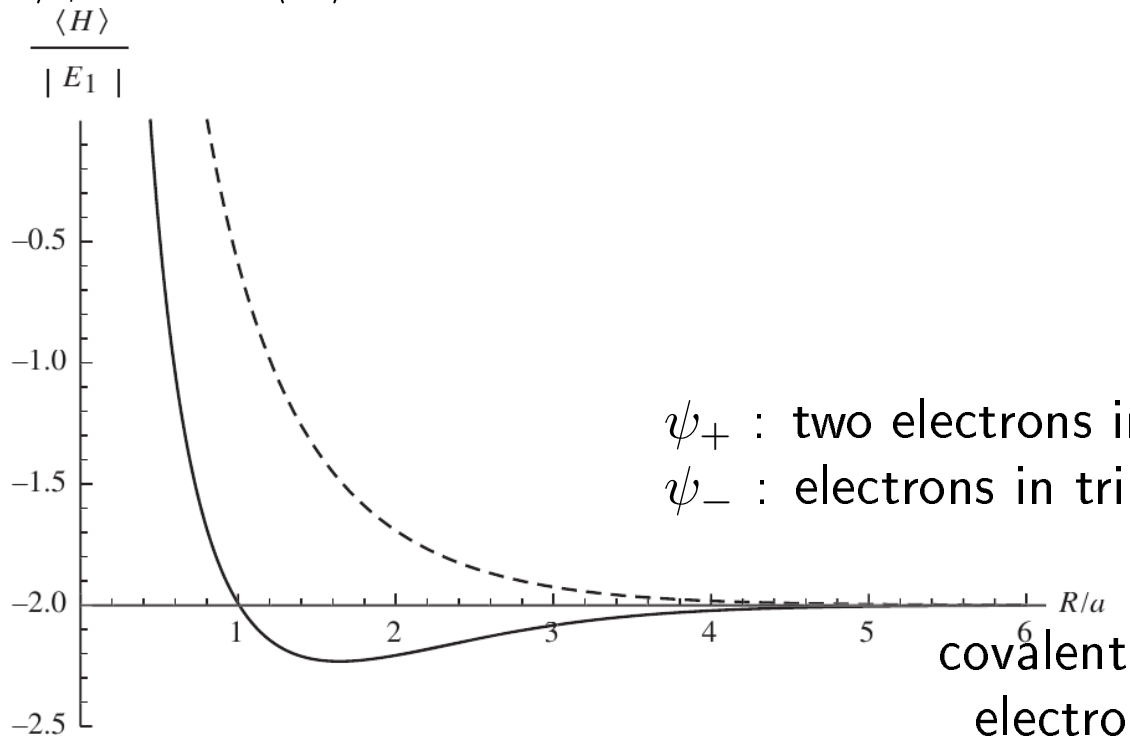
ψ_- : electrons in triplet spin configuration

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Locating the minimum on the plot, bond length $= 1.64$ a.u.; expt. : 1.40 a.u.

binding energy $= 3.15$ eV

$\langle H \rangle_+$ versus $\langle H \rangle_-$



ψ_+ : two electrons in singlet spin configuration

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Locating the minimum on the plot, bond length = 1.64 a.u.; expt. : 1.40 a.u.

binding energy = 3.15 eV expt. : 4.75 eV

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$\Longrightarrow$ strong preference for having the electron spins anti-aligned

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—but the exchange splitting isn't

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H_2 is a sort of inchoate anti-ferromagnet where the Hamiltonian, independent of the spin, selects a certain spatial ground state and spin state comes along for the ride, to satisfy the Fermi statistics .